

HIGHER EDUCATION STUDENTS PERFORMANCE EVALUATION

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1	STUDENTID	age	gender	high schools type	scholarship type	work	sports activity	total salary	weekly study hours	reading freq.(non-scientific books/journals)	reading freq.(scientific books)	attendance to the seminars	attendance to classes	preparation to exam	taking notes	listening in classes	discussions	cumulative GPA	expected GPA	output grades
2	STUDENT1	22-25	male	other	50%	yes	no	\$ 135-200	6 - 10 hours	sometimes	sometimes	yes	always	closest date to the exam	always	sometimes	never	< 2	< 2	poor
3	STUDENT2	22-25	male	other	50%	yes	no	\$ 135-200	< 5 hours	sometimes	sometimes	yes	always	closest date to the exam	always	sometimes	always	2 - 2.49	2.50 - 2.99	poor
4	STUDENT3	22-25	male	state	50%	no	no	\$ 201-270	< 5 hours	none	sometimes	yes	always	closest date to the exam	sometimes	sometimes	never	2 - 2.49	2 - 2.49	poor
5	STUDENT4	18-21	female	private	50%	yes	no	\$ 201-270	6 - 10 hours	none	sometimes	yes	always	regularly during the semester	always	sometimes	sometimes	2.50 - 2.99	2 - 2.49	poor
6	STUDENT5	22-25	male	private	50%	no	no	\$ 271-340	< 5 hours	none	none	yes	always	closest date to the exam	sometimes	sometimes	sometimes	2 - 2.49	2 - 2.49	poor
7	STUDENT6	22-25	male	state	50%	no	no	\$ 201-270	none	none	sometimes	yes	always	closest date to the exam	sometimes	sometimes	never	3.00 - 3.49	3.00 - 3.49	poor to satisfactory
8	STUDENT7	18-21	male	state	75%	no	no	\$ 135-200	< 5 hours	sometimes	sometimes	no	sometimes	closest date to the exam	always	always	always	3.00 - 3.49	3.00 - 3.49	good
9	STUDENT8	18-21	female	state	50%	yes	yes	\$ 201-270	none	sometimes	sometimes	yes	always	closest date to the exam	always	sometimes	sometimes	< 2	< 2	poor to satisfactory
10	STUDENT9	22-25	female	other	50%	no	yes	\$ 135-200	none	sometimes	sometimes	yes	always	closest date to the exam	always	sometimes	sometimes	3.00 - 3.49	2.50 - 2.99	good
11	STUDENT10	22-25	female	state	50%	no	no	\$ 271-340	< 5 hours	sometimes	sometimes	yes	sometimes	closest date to the exam	sometimes	sometimes	sometimes	< 2	2 - 2.49	fail
12	STUDENT11	18-21	female	private	50%	no	no	\$ 271-340	none	none	none	yes	sometimes	closest date to the exam	sometimes	sometimes	sometimes	< 2	< 2	poor to satisfactory
13	STUDENT12	18-21	female	private	75%	yes	yes	\$ 341-410	6 - 10 hours	often	often	yes	always	regularly during the semester	always	never	always	3.00 - 3.49	2.50 - 2.99	fail
14	STUDENT13	18-21	female	private	75%	no	no	\$ 135-200	6 - 10 hours	sometimes	sometimes	yes	always	closest date to the exam	sometimes	sometimes	sometimes	3.00 - 3.49	2 - 2.49	fail
15	STUDENT14	22-25	female	state	full	no	no	\$ 135-200	none	sometimes	none	yes	sometimes	closest date to the exam	always	sometimes	always	3.00 - 3.49	2 - 2.49	poor
16	STUDENT15	above 26	male	state	75%	yes	yes	\$ 271-340	< 5 hours	sometimes	sometimes	yes	sometimes	closest date to the exam	sometimes	always	sometimes	3.00 - 3.49	3.00 - 3.49	poor to satisfactory
17	STUDENT16	22-25	male	state	50%	no	no	\$ 135-200	< 5 hours	sometimes	sometimes	no	sometimes	closest date to the exam	always	sometimes	sometimes	2 - 2.49	2 - 2.49	poor to satisfactory
18	STUDENT17	18-21	female	state	full	no	yes	\$ 135-200	< 5 hours	sometimes	sometimes	yes	sometimes	closest date to the exam	always	sometimes	always	3.00 - 3.49	2.50 - 2.99	poor
19	STUDENT18	22-25	male	state	50%	no	no	\$ 135-200	< 5 hours	sometimes	sometimes	no	sometimes	closest date to the exam	sometimes	sometimes	sometimes	2 - 2.49	2 - 2.49	poor to satisfactory
20	STUDENT19	18-21	female	state	75%	no	no	\$ 271-340	more than 20 hours	often	sometimes	no	sometimes	closest date to the exam	always	never	always	2.50 - 2.99	2.50 - 2.99	poor to satisfactory
21	STUDENT20	18-21	male	private	50%	no	no	\$ 201-270	11 - 20 hours	sometimes	sometimes	yes	always	closest date to the exam	always	sometimes	sometimes	2 - 2.49	2.50 - 2.99	satisfactory
22	STUDENT21	18-21	male	state	full	yes	no	\$ 135-200	11 - 20 hours	sometimes	sometimes	no	always	regularly during the semester	always	never	sometimes	3.00 - 3.49	3.00 - 3.49	poor
23	STUDENT22	18-21	male	state	full	no	no	\$ 135-200	6 - 10 hours	sometimes	sometimes	yes	always	closest date to the exam	always	never	always	2.50 - 2.99	2.50 - 2.99	poor
24	STUDENT23	22-25	male	state	50%	yes	no	\$ 135-200	6 - 10 hours	often	often	yes	always	regularly during the semester	always	never	sometimes	2.50 - 2.99	2.50 - 2.99	satisfactory
25	STUDENT24	above 26	male	state	25%	yes	yes	above \$ 410	6 - 10 hours	sometimes	often	yes	sometimes	closest date to the exam	always	sometimes	always	2.50 - 2.99	2.50 - 2.99	poor
26	STUDENT25	22-25	male	state	50%	no	no	\$ 201-270	none	sometimes	sometimes	yes	always	closest date to the exam	sometimes	never	always	3.00 - 3.49	3.00 - 3.49	poor to satisfactory
27	STUDENT26	22-25	male	state	50%	no	no	\$ 135-200	< 5 hours	sometimes	sometimes	yes	always	closest date to the exam	sometimes	never	always	< 2	2 - 2.49	poor to satisfactory
28	STUDENT27	22-25	male	state	50%	no	yes	\$ 135-200	< 5 hours	often	sometimes	yes	always	closest date to the exam	always	always	always	2 - 2.49	< 2	poor
29	STUDENT28	18-21	male	private	50%	yes	no	\$ 135-200	6 - 10 hours	sometimes	sometimes	yes	sometimes	closest date to the exam	always	never	sometimes	2 - 2.49	< 2	poor
30	STUDENT29	above 26	male	state	50%	no	no	\$ 135-200	< 5 hours	sometimes	sometimes	yes	always	closest date to the exam	always	sometimes	always	> 3.49	3.00 - 3.49	satisfactory
31	STUDENT30	22-25	male	other	75%	no	no	\$ 135-200	< 5 hours	sometimes	sometimes	yes	always	closest date to the exam	always	sometimes	sometimes	3.00 - 3.49	2.50 - 2.99	good
32	STUDENT31	22-25	male	state	full	yes	yes	\$ 135-200	< 5 hours	sometimes	sometimes	yes	always	closest date to the exam	sometimes	always	always	> 3.49	3.00 - 3.49	good
33	STUDENT32	above 26	male	state	50%	yes	no	\$ 135-200	< 5 hours	sometimes	often	yes	always	closest date to the exam	always	always	sometimes	3.00 - 3.49	2.50 - 2.99	satisfactory
34	STUDENT33	22-25	female	state	50%	no	no	\$ 201-270	< 5 hours	sometimes	sometimes	yes	sometimes	closest date to the exam	sometimes	sometimes	sometimes	2 - 2.49	2.50 - 2.99	poor
35	STUDENT34	22-25	female	state	50%	yes	no	\$ 135-200	< 5 hours	none	sometimes	yes	always	closest date to the exam	never	always	sometimes	2 - 2.49	2.50 - 2.99	poor to satisfactory
36	STUDENT35	18-21	male	private	50%	no	no	\$ 201-270	< 5 hours	none	none	yes	sometimes	closest date to the exam	sometimes	sometimes	always	3.00 - 3.49	< 2	poor to satisfactory
37	STUDENT36	18-21	male	private	75%	no	no	\$ 271-340	< 5 hours	sometimes	sometimes	yes	sometimes	closest date to the exam	sometimes	never	always	< 2	< 2	poor
38	STUDENT37	22-25	male	other	75%	yes	no	\$ 271-340	none	none	often	yes	sometimes	closest date to the exam	sometimes	never	sometimes	3.00 - 3.49	2.50 - 2.99	poor to satisfactory
39	STUDENT38	22-25	male	state	50%	yes	yes	\$ 201-270	< 5 hours	sometimes	sometimes	yes	always	closest date to the exam	always	sometimes	always	> 3.49	3.00 - 3.49	poor
40	STUDENT39	22-25	male	state	full	no	no	\$ 135-200	6 - 10 hours	sometimes	sometimes	yes	always	closest date to the exam	sometimes	sometimes	sometimes	3.00 - 3.49	2.50 - 2.99	poor to satisfactory
41	STUDENT40	22-25	female	state	50%	no	no	\$ 135-200	none	sometimes	none	yes	always	closest date to the exam	sometimes	sometimes	sometimes	2 - 2.49	< 2	poor
42	STUDENT41	18-21	male	private	50%	no	no	\$ 201-270	none	sometimes	sometimes	yes	always	closest date to the exam	always	never	sometimes	2 - 2.49	2.50 - 2.99	poor
43	STUDENT42	above 26	male	state	50%	yes	no	\$ 201-270	6 - 10 hours	sometimes	sometimes	yes	sometimes	closest date to the exam	always	sometimes	sometimes	3.00 - 3.49	2.50 - 2.99	poor
44	STUDENT43	22-25	male	state	50%	no	yes	\$ 135-200	< 5 hours	sometimes	none	yes	always	closest date to the exam	sometimes	never	always	2 - 2.49	2.50 - 2.99	poor
45	STUDENT44	18-21	male	state	50%	no	no	\$ 135-200	< 5 hours	sometimes	sometimes	yes	sometimes	closest date to the exam	always	never	always	2.50 - 2.99	2 - 2.49	satisfactory to good
46	STUDENT45	22-25	male	other	50%	no	no	\$ 135-200	< 5 hours	sometimes	sometimes	yes	always	closest date to the exam	always	sometimes	sometimes	3.00 - 3.49	2.50 - 2.99	poor
47	STUDENT46	18-21	male	state	50%	no	no	\$ 341-410	6 - 10 hours	sometimes	sometimes	yes	always	closest date to the exam	sometimes	sometimes	sometimes	3.00 - 3.49	2.50 - 2.99	satisfactory
48	STUDENT47	22-25	male	state	50%	no	no	\$ 135-200	< 5 hours	sometimes	sometimes	yes	always	closest date to the exam	sometimes	sometimes	always	3.00 - 3.49	2 - 2.49	good
49	STUDENT48	22-25	male	state	50%	no	no	\$ 135-200	< 5 hours	sometimes	sometimes	yes	sometimes	closest date to the exam	sometimes	never	always	> 3.49	2.50 - 2.99	satisfactory
50	STUDENT49	18-21	male	state	50%	no	yes	\$ 135-200	none	none	sometimes	yes	always	closest date to the exam	always	sometimes	always	2 - 2.49	2 - 2.49	poor
51	STUDENT50	18-21	male	private	75%	no	no	\$ 201-270	< 5 hours	sometimes	sometimes	yes	always	closest date to the exam	sometimes	sometimes	sometimes	2.50 - 2.99	2 - 2.49	poor to satisfactory
52	STUDENT51	22-25	male	state	50%	no	no	\$ 201-270	< 5 hours	sometimes	sometimes	yes	sometimes	closest date to the exam	sometimes	sometimes	sometimes	2.50 - 2.99	2.50 - 2.99	poor
53	STUDENT52	22-25	female	other	50%	yes	yes	\$ 135-200	none	sometimes	sometimes	yes	sometimes	closest date to the exam	sometimes	always	always	2.50 - 2.99	2.50 - 2.99	satisfactory to good
54	STUDENT53	22-25	female	state	50%	yes	no	\$ 201-270	6 - 10 hours	none	always	yes	always	regularly during the semester	sometimes	sometimes	sometimes	3.00 - 3.49	2 - 2.49	poor
55	STUDENT54	22-25	female	state	50%	no	yes	\$ 201-270	none	sometimes	sometimes	yes	sometimes	closest date to the exam	sometimes	sometimes	sometimes	2.50 - 2.99	2 - 2.49	good
56	STUDENT55	22-25	male	state	50%	no	no	\$ 271-340	< 5 hours	none	none	yes	always	regularly during the semester	always	never	always	> 3.49	2.50 - 2.99	satisfactory
57	STUDENT56	above 26	male	state	50%	yes	no	\$ 341-410	none	sometimes	sometimes	yes	always	never	never	always	always	> 3.49	3.00 - 3.49	satisfactory
58	STUDENT57	22-25	male	state	50%	no	yes	\$ 135-200	11 - 20 hours	sometimes	often	yes	always	regularly during the semester	always	sometimes	always	> 3.49	3.00 - 3.49	good
59	STUDENT58	22-25	male	state	50%	yes	yes	\$ 135-200	< 5 hours	often	sometimes	yes	always	closest date to the exam	always	sometimes	always	> 3.49	3.00 - 3.49	satisfactory to good
60	STUDENT59	above 26	male	state	50%	no	no	\$ 271-340	6 - 10 hours	sometimes	sometimes	yes	always	closest date to the exam	always	sometimes	sometimes	> 3.49	3.00 - 3.49	satisfactory
61	STUDENT60	22-25	male	state	50%	no	yes	\$ 135-200	6 - 10 hours	sometimes	sometimes	yes	always	closest date to the exam	always	always	sometimes	3.00 - 3.49	2.50 - 2.99	good
62	STUDENT61	22-25	female	state	50%	no	no	above \$ 410	none	sometimes	sometimes	yes	always	closest date to the exam	never	always	always	2 - 2.49	< 2	poor to satisfactory
63	STUDENT62	18-21	male	other	50%	no	yes	\$ 201-270	6 - 10 hours	none	sometimes	yes	sometimes	closest date to the exam	always	sometimes	sometimes	3.00 - 3.49	3.00 - 3.49	good
64	STUDENT63	22-25	male	state	50%	no	no	\$ 135-200	6 - 10 hours	sometimes	sometimes	yes	always	closest date to the exam	sometimes	always	always	> 3.49	3.00 - 3.49	satisfactory
65	STUDENT64	22-25	male	state	75%	no	no	\$ 201-270	none	none	sometimes	yes	always	closest date to the exam	sometimes	sometimes	sometimes	2.50 - 2.99	2.50 - 2.99	good
66	STUDENT65	22-25	male	other	full	no	no	\$ 135-200	none	sometimes	sometimes	yes	always	closest date to the exam	always	sometimes	always	3.00 - 3.49	2.50 - 2.99	satisfactory
67	STUDENT66	18-21	male	state	50%	no	no	\$ 271-340	none	none	sometimes	yes	sometimes	closest date to the exam	sometimes	never	sometimes	2 - 2.49	2 - 2.49	poor to satisfactory
68	STUDENT67	22-25	male	state	50%	no	no	\$ 135-200	< 5 hours	sometimes	sometimes	yes	always	closest date to the exam	always	sometimes	sometimes	> 3.49	3.00 - 3.49	good
69	STUDENT68	22-25	male	other	50%	yes	yes	\$ 271-340	< 5 hours	sometimes	sometimes	yes	sometimes	closest date to the exam	sometimes	sometimes	never	2 - 2.49	2.50 - 2.99	poor
70	STUDENT69	22-25	female	state	75%	yes	no	\$ 135-200	none	sometimes	sometimes	yes	sometimes	closest date to the exam	sometimes	sometimes	always	3.00 - 3.49	2.50 - 2.99	good
71	STUDENT70	22-25	female	state	75%	no	no	\$ 135-200	none	none	sometimes	yes	always	closest date to the exam	always	sometimes	always	2.50 - 2.99	2 - 2.49	good
72	STUDENT71	18-21	male	state	75%	no	yes	\$ 135-200	< 5 hours	sometimes	sometimes	yes	always	closest date to the exam	sometimes	sometimes	always	> 3.49	3.00 - 3.49	excellent
73	STUDENT72	18-21	female	other	75%	no	no	\$ 135-200	< 5 hours	sometimes	none	yes	always	closest date to the exam	sometimes	always	always	2 - 2.49	2 - 2.49	good to excellent
74	STUDENT73	18-21	male																	

82	STUDENT81	22-25	female	state	50%	yes	no	\$ 135-200	none	none	none	yes	always	closest date to the exam	sometimes	always	never	3.00 - 3.49	2 - 2.49	satisfactory to good
83	STUDENT82	above 26	male	state	50%	yes	no	\$ 135-200	< 5 hours	sometimes	sometimes	yes	always	closest date to the exam	always	always	sometimes	> 3.49	3.00 - 3.49	satisfactory
84	STUDENT83	22-25	male	state	75%	yes	no	\$ 201-270	< 5 hours	sometimes	often	yes	always	closest date to the exam	always	always	sometimes	3.00 - 3.49	2.50 - 2.99	excellent
85	STUDENT84	22-25	male	other	50%	no	no	\$ 271-340	none	sometimes	sometimes	yes	always	closest date to the exam	always	sometimes	always	< 2	2 - 2.49	excellent
86	STUDENT85	above 26	male	other	50%	yes	no	\$ 271-340	none	sometimes	sometimes	yes	always	never	always	always	always	> 3.49	3.00 - 3.49	excellent
87	STUDENT86	18-21	male	state	full	no	no	\$ 201-270	none	none	none	yes	always	closest date to the exam	always	sometimes	always	3.00 - 3.49	2.50 - 2.99	satisfactory to good
88	STUDENT87	22-25	male	state	75%	no	no	\$ 135-200	< 5 hours	sometimes	sometimes	yes	always	closest date to the exam	always	sometimes	always	> 3.49	3.00 - 3.49	good
89	STUDENT88	22-25	male	state	50%	no	yes	\$ 201-270	< 5 hours	sometimes	sometimes	yes	always	closest date to the exam	sometimes	sometimes	sometimes	< 2	< 2	good to excellent
90	STUDENT89	18-21	male	state	75%	no	yes	\$ 135-200	< 5 hours	sometimes	sometimes	yes	always	regularly during the semester	always	never	sometimes	> 3.49	2.50 - 2.99	good to excellent
91	STUDENT90	22-25	male	state	50%	no	no	\$ 135-200	< 5 hours	sometimes	sometimes	yes	sometimes	closest date to the exam	always	never	always	2.50 - 2.99	2 - 2.49	good to excellent
92	STUDENT91	22-25	female	state	50%	no	yes	\$ 135-200	< 5 hours	sometimes	sometimes	yes	always	closest date to the exam	sometimes	always	sometimes	3.00 - 3.49	2 - 2.49	good to excellent
93	STUDENT92	22-25	male	state	full	yes	yes	\$ 135-200	< 5 hours	sometimes	sometimes	yes	sometimes	closest date to the exam	sometimes	always	sometimes	> 3.49	3.00 - 3.49	good to excellent
94	STUDENT93	18-21	male	state	50%	no	no	\$ 135-200	6 - 10 hours	sometimes	sometimes	yes	always	closest date to the exam	always	sometimes	always	2 - 2.49	2 - 2.49	excellent
95	STUDENT94	18-21	male	state	none	yes	no	\$ 135-200	more than 20 hours	sometimes	none	yes	always	closest date to the exam	always	always	always	2 - 2.49	2 - 2.49	satisfactory to good
96	STUDENT95	22-25	male	state	50%	no	no	\$ 135-200	< 5 hours	sometimes	often	yes	always	closest date to the exam	always	sometimes	sometimes	> 3.49	3.00 - 3.49	good to excellent
97	STUDENT96	18-21	male	other	full	yes	yes	\$ 135-200	< 5 hours	sometimes	often	yes	always	closest date to the exam	always	sometimes	sometimes	3.00 - 3.49	2.50 - 2.99	good
98	STUDENT97	18-21	male	state	75%	no	yes	\$ 135-200	< 5 hours	often	sometimes	yes	always	closest date to the exam	always	sometimes	sometimes	2 - 2.49	2.50 - 2.99	excellent
99	STUDENT98	18-21	male	state	75%	yes	no	\$ 135-200	< 5 hours	sometimes	sometimes	yes	always	closest date to the exam	always	sometimes	sometimes	2.50 - 2.99	2.50 - 2.99	good to excellent
100	STUDENT99	18-21	male	state	75%	no	yes	\$ 135-200	< 5 hours	often	sometimes	yes	always	closest date to the exam	always	never	sometimes	3.00 - 3.49	3.00 - 3.49	excellent
101	STUDENT100	22-25	male	state	50%	no	no	\$ 135-200	6 - 10 hours	sometimes	sometimes	yes	always	closest date to the exam	always	sometimes	always	2 - 2.49	2 - 2.49	excellent
102	STUDENT101	18-21	male	state	75%	no	no	\$ 135-200	< 5 hours	sometimes	sometimes	yes	always	closest date to the exam	sometimes	sometimes	sometimes	2.50 - 2.99	2.50 - 2.99	good to excellent
103	STUDENT102	18-21	male	state	75%	no	no	\$ 201-270	< 5 hours	sometimes	always	yes	always	closest date to the exam	always	always	sometimes	2 - 2.49	2.50 - 2.99	excellent
104	STUDENT103	18-21	male	state	50%	no	no	\$ 135-200	< 5 hours	sometimes	sometimes	no	always	closest date to the exam	always	never	sometimes	2.50 - 2.99	3.00 - 3.49	excellent
105	STUDENT104	18-21	male	private	75%	no	no	above \$ 410	< 5 hours	often	sometimes	yes	always	closest date to the exam	sometimes	sometimes	always	3.00 - 3.49	3.00 - 3.49	excellent
106	STUDENT105	18-21	female	state	50%	no	no	\$ 135-200	< 5 hours	sometimes	sometimes	yes	always	closest date to the exam	sometimes	always	sometimes	< 2	< 2	satisfactory
107	STUDENT106	18-21	male	state	75%	yes	yes	\$ 135-200	< 5 hours	often	sometimes	yes	sometimes	regularly during the semester	sometimes	never	always	2.50 - 2.99	2.50 - 2.99	excellent
108	STUDENT107	18-21	male	state	75%	no	yes	\$ 135-200	6 - 10 hours	sometimes	sometimes	yes	sometimes	regularly during the semester	sometimes	sometimes	sometimes	3.00 - 3.49	3.00 - 3.49	excellent
109	STUDENT108	18-21	female	state	75%	yes	yes	\$ 135-200	< 5 hours	sometimes	sometimes	yes	always	closest date to the exam	sometimes	always	sometimes	3.00 - 3.49	2 - 2.49	good to excellent
110	STUDENT109	22-25	female	private	full	no	yes	\$ 201-270	6 - 10 hours	often	sometimes	yes	sometimes	closest date to the exam	sometimes	always	never	2.50 - 2.99	2.50 - 2.99	good to excellent
111	STUDENT110	18-21	male	private	50%	no	yes	\$ 135-200	6 - 10 hours	sometimes	sometimes	yes	always	regularly during the semester	always	sometimes	always	3.00 - 3.49	3.00 - 3.49	excellent
112	STUDENT111	22-25	male	state	50%	no	no	\$ 201-270	< 5 hours	sometimes	sometimes	yes	sometimes	closest date to the exam	sometimes	sometimes	sometimes	2 - 2.49	2.50 - 2.99	poor to satisfactory
113	STUDENT112	18-21	female	private	full	no	yes	\$ 135-200	none	sometimes	sometimes	no	always	closest date to the exam	always	never	always	3.00 - 3.49	2.50 - 2.99	poor to satisfactory
114	STUDENT113	22-25	female	other	50%	yes	no	\$ 135-200	none	none	none	no	always	closest date to the exam	always	always	always	3.00 - 3.49	2 - 2.49	poor to satisfactory
115	STUDENT114	22-25	female	other	50%	no	no	\$ 135-200	< 5 hours	sometimes	often	no	sometimes	closest date to the exam	sometimes	sometimes	never	< 2	2 - 2.49	poor
116	STUDENT115	22-25	female	state	full	yes	yes	\$ 341-410	< 5 hours	sometimes	sometimes	no	sometimes	closest date to the exam	always	sometimes	never	< 2	< 2	poor to satisfactory
117	STUDENT116	22-25	female	state	full	yes	no	\$ 135-200	6 - 10 hours	often	sometimes	no	always	regularly during the semester	always	sometimes	always	2 - 2.49	2 - 2.49	poor
118	STUDENT117	22-25	female	state	full	no	no	\$ 135-200	6 - 10 hours	often	often	no	always	closest date to the exam	always	always	always	2 - 2.49	2.50 - 2.99	poor
119	STUDENT118	above 26	female	private	50%	no	yes	\$ 135-200	< 5 hours	sometimes	often	no	sometimes	closest date to the exam	sometimes	always	sometimes	2 - 2.49	2.50 - 2.99	poor
120	STUDENT119	18-21	male	state	full	no	yes	\$ 135-200	none	sometimes	sometimes	yes	always	closest date to the exam	always	always	sometimes	2.50 - 2.99	2.50 - 2.99	poor
121	STUDENT120	22-25	female	state	75%	no	yes	\$ 135-200	more than 20 hours	sometimes	always	yes	sometimes	regularly during the semester	sometimes	sometimes	sometimes	2.50 - 2.99	2.50 - 2.99	poor to satisfactory
122	STUDENT121	22-25	female	private	50%	yes	yes	\$ 201-270	none	none	none	yes	sometimes	closest date to the exam	always	always	always	2 - 2.49	2 - 2.49	poor
123	STUDENT122	22-25	female	state	50%	yes	yes	\$ 135-200	more than 20 hours	often	always	yes	never	closest date to the exam	never	never	never	< 2	< 2	fail
124	STUDENT123	18-21	male	state	full	no	yes	\$ 135-200	< 5 hours	none	sometimes	yes	always	closest date to the exam	always	sometimes	sometimes	3.00 - 3.49	2.50 - 2.99	poor to satisfactory
125	STUDENT124	22-25	female	other	50%	no	no	\$ 271-340	< 5 hours	sometimes	often	no	always	closest date to the exam	always	always	sometimes	< 2	< 2	poor
126	STUDENT125	18-21	female	state	75%	yes	yes	\$ 135-200	11 - 20 hours	sometimes	sometimes	no	always	regularly during the semester	always	always	sometimes	2.50 - 2.99	2.50 - 2.99	satisfactory
127	STUDENT126	18-21	female	state	full	yes	yes	\$ 135-200	6 - 10 hours	sometimes	often	no	always	regularly during the semester	always	sometimes	always	< 2	2.50 - 2.99	poor to satisfactory
128	STUDENT127	18-21	female	private	75%	yes	yes	\$ 271-340	11 - 20 hours	sometimes	often	yes	always	closest date to the exam	always	always	always	2 - 2.49	2 - 2.49	satisfactory
129	STUDENT128	18-21	female	state	75%	no	no	\$ 135-200	< 5 hours	none	none	no	always	closest date to the exam	always	sometimes	sometimes	2 - 2.49	2 - 2.49	poor
130	STUDENT129	18-21	female	state	75%	no	yes	\$ 135-200	< 5 hours	none	none	no	always	closest date to the exam	always	sometimes	sometimes	2 - 2.49	2 - 2.49	fail
131	STUDENT130	18-21	female	state	50%	yes	yes	\$ 135-200	11 - 20 hours	sometimes	often	no	always	regularly during the semester	always	sometimes	always	2 - 2.49	2.50 - 2.99	satisfactory
132	STUDENT131	18-21	female	state	50%	yes	yes	\$ 135-200	11 - 20 hours	sometimes	sometimes	no	always	regularly during the semester	always	never	always	2 - 2.49	2.50 - 2.99	poor
133	STUDENT132	18-21	female	private	full	no	yes	\$ 135-200	11 - 20 hours	none	often	no	always	regularly during the semester	always	sometimes	always	> 3.49	2.50 - 2.99	satisfactory to good
134	STUDENT133	18-21	female	private	full	no	yes	\$ 135-200	< 5 hours	sometimes	often	no	always	closest date to the exam	always	sometimes	always	> 3.49	3.00 - 3.49	satisfactory
135	STUDENT134	18-21	female	state	full	no	no	\$ 135-200	6 - 10 hours	none	sometimes	no	always	closest date to the exam	always	always	always	> 3.49	2.50 - 2.99	satisfactory
136	STUDENT135	18-21	female	state	75%	no	yes	\$ 135-200	< 5 hours	sometimes	sometimes	no	always	closest date to the exam	sometimes	never	sometimes	2 - 2.49	2.50 - 2.99	poor
137	STUDENT136	22-25	female	state	50%	yes	yes	\$ 135-200	< 5 hours	often	sometimes	yes	always	closest date to the exam	always	always	sometimes	2.50 - 2.99	2 - 2.49	poor to satisfactory
138	STUDENT137	18-21	female	state	50%	yes	yes	\$ 135-200	< 5 hours	sometimes	sometimes	no	always	closest date to the exam	always	sometimes	sometimes	2 - 2.49	2 - 2.49	fail
139	STUDENT138	18-21	female	private	full	no	yes	\$ 135-200	< 5 hours	sometimes	sometimes	yes	always	closest date to the exam	always	never	always	2 - 2.49	3.00 - 3.49	poor to satisfactory
140	STUDENT139	18-21	female	state	75%	yes	yes	above \$ 410	< 5 hours	sometimes	often	no	always	closest date to the exam	always	sometimes	always	< 2	2.50 - 2.99	fail
141	STUDENT140	18-21	female	state	75%	no	yes	\$ 201-270	none	none	sometimes	no	sometimes	closest date to the exam	sometimes	always	sometimes	< 2	2 - 2.49	fail
142	STUDENT141	22-25	female	state	50%	yes	yes	\$ 135-200	6 - 10 hours	often	sometimes	yes	always	closest date to the exam	sometimes	never	sometimes	2.50 - 2.99	2.50 - 2.99	good
143	STUDENT142	18-21	female	state	75%	no	no	\$ 135-200	6 - 10 hours	sometimes	sometimes	no	sometimes	closest date to the exam	always	sometimes	sometimes	> 3.49	2.50 - 2.99	good
144	STUDENT143	18-21	female	private	75%	no	no	\$ 135-200	< 5 hours	sometimes	sometimes	yes	always	closest date to the exam	sometimes	always	sometimes	3.00 - 3.49	2.50 - 2.99	poor
145	STUDENT144	22-25	female	state	75%	yes	yes	above \$ 410	< 5 hours	sometimes	none	yes	always	closest date to the exam	sometimes	never	sometimes	> 3.49	2.50 - 2.99	satisfactory to good
146	STUDENT145	18-21	female	private	full	no	no	\$ 271-340	6 - 10 hours	none	none	yes	always	closest date to the exam	always	sometimes	always	> 3.49	3.00 - 3.49	satisfactory

WHY WE CHOSE THIS DATA SET?

We chose this data set because the "Higher Education Students Performance Evaluation" dataset provides an abundant amount of data for researching and comprehending student performance in higher education, helping researchers and educators to make data-driven decisions and acquire understanding of the variables that influence students' success.





01

STRATIFIED RANDOM SAMPLING

STRATIFIED RANDOM SAMPLING

```
# Create a new data frame
new_weekly_data <- data.frame("intervals" = c("0 - 10 hours", "11 - 20 hours", "20 - 30 hours")
                             "frequencies" = c(sum(weekly_data$frequencies[1:3]),
                                                weekly_data$frequencies[4], weekly_data$frequencies[5]))

# Print the new data frame
print(new_weekly_data)

#We create a new column with the midpoints for each interval
xi_midpoint <- c(5,15.5,25)
new_weekly_data$xi <- xi_midpoint

#We calculate the mean for the grouped data
xbar=sum(new_weekly_data$xi*new_weekly_data$frequencies)/sum(new_weekly_data$frequencies)
print (xbar)

#We calculate the variance for the grouped data
variance <- sum((new_weekly_data$xi-xbar)^2*new_weekly_data$frequencies)/n
print(variance)

#We calculate the standard deviation for the grouped data
s <- sqrt(variance)
print (s)
```

We calculate the mean for the grouped data and as a result, we obtain that on average, the students would spend 6.13 hours for weekly studying.

The variance value is 15.838 and the standard deviation is 3.979.

A standard deviation of 3.979 indicates the average amount of dispersion or spread of the data points from the mean. It suggests that, on average, the data points in our dataset deviate from the mean by approximately 3.979 units.

```

#We compute the confidence interval for 95% confidence level
z <- 2
#We compute the margin of error
margin <- z*s/sqrt(n)
print (margin)

# Confidence interval for the population mean
lowerinterval <- xbar - margin
print(lowerinterval)

upperinterval <- xbar + margin
print(upperinterval)

```

```

> print (margin)
[1] 1.198048

> print(lowerinterval)
[1] 4.932987

> print(upperinterval)
[1] 7.329082

```

We take the z value equal to 2 at a 95% confidence level and we get the margin of error as being 1.198. The confidence intervals for the population mean has the lower limit of 4.932 and the upper one 7.329. About 95% of the observations from the entire population has the mean between 4.932 and 7.329.

We take the z value equal to 2.6 at a 99% confidence level and we get the margin of error as 1.557. Therefore, the lower limit for the population mean is 4.573 and the upper one is 7.688. This confidence interval applies in 99% of the cases.

```

#We compute the confidence interval for 99% confidence level
z1 <- 2.6
#We compute the margin of error
margin1 <- z1*s/sqrt(n)

# Confidence interval for the population mean
lowerinterval1 <- xbar - margin1
print(lowerinterval1)

upperinterval1 <- xbar + margin1
print(upperinterval1)

```

```

> print(margin1)
[1] 1.557462

> print(lowerinterval1)
[1] 4.573572

> print(upperinterval1)
[1] 7.688497

```



02

HYPOTHESIS TESTING

HYPOTESIS TESTING-ONE TAILED TEST



LOWER TAIL TEST

```
xbar_sample <- 6.40 #mean for sample
mu0 <- 10          # hypothesized value
sample_s <- 4.49   #standard deviation for sample
n_sample <- 80     # sample size
t <- (xbar_sample-mu0)/(sample_s/sqrt(n_sample))
print (t)          #test statistics

alpha = 0.05       #We compute the critical value at 0.05 significance level
t.alpha=qt(1-alpha, df=n_sample-1)
-t.alpha           #critical value
```

H0: $\mu \geq 10$
H1: $\mu < 10$

> print (t)
[1] -7.171354

> -t.alpha
[1] -1.664371

The test statistic -7.171354 is less than the critical value of -1.664371. Hence, at 0.05 significance level, we can reject the claim that the average weekly study hours for a student with a job is more than 10 hours.

HYPOTESIS TESTING-ONE TAILED TEST



```
xbar_sample <- 6.40 #mean for sample
mu0 <- 13          # hypothesized value
sample_s <- 4.49   #standard deviation for sample
n_sample <- 80     # sample size
t <- (xbar_sample-mu0)/(sample_s/sqrt(n_sample))
print (t)          #test statistics

alpha = 0.05       #We compute the critical value at 0.05 significance level
t.alpha=qt(1-alpha, df=n_sample-1)
t.alpha           #critical value
```

H0: $\mu \leq 13$
H1: $\mu < 13$

```
> print (t)
[1] -13.14748

> t.alpha
[1] 1.664371
```

The test statistic -13.14748 is less than the critical value 1.664371. So, at 0.05 significance level, we can accept the claim that the average weekly study hours for students with a job is less than 13 hours

UPPER
TAIL
TEST

HYPOTESIS TESTING-TWO TAILED TEST

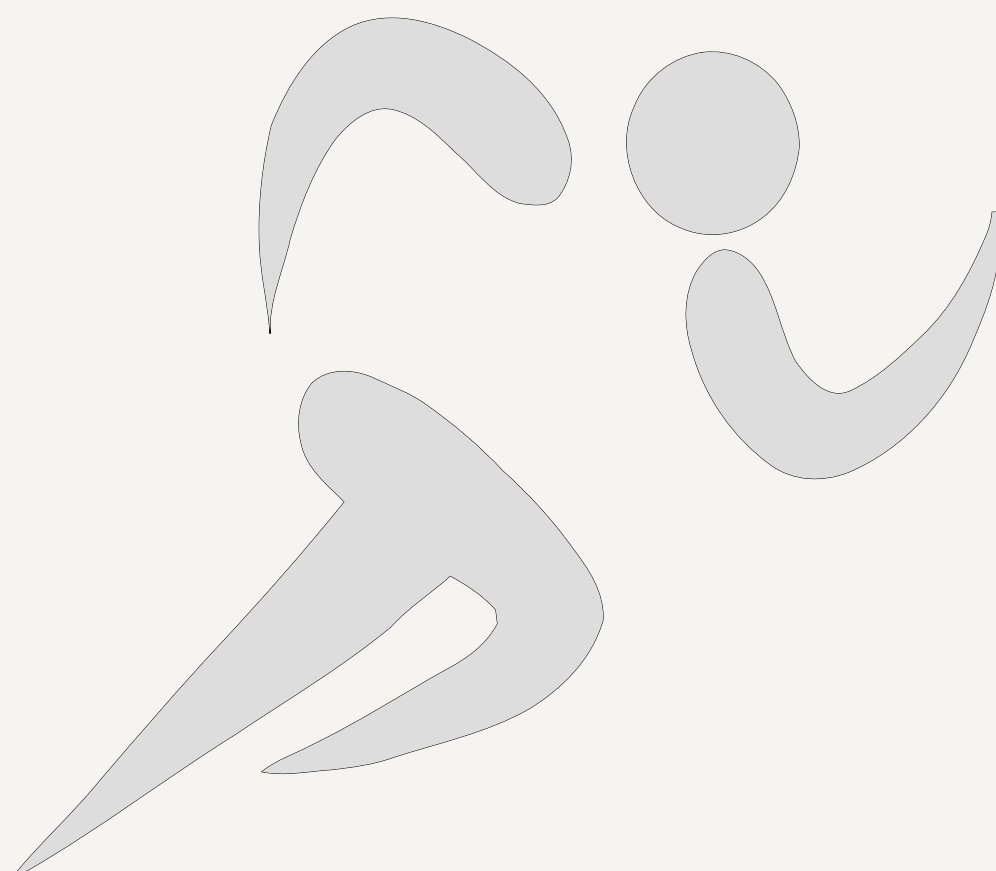
```
xbar_sample <- 6.40           # sample mean
mu0 <- 11.5                   # hypothesized value
sample_s <- 4.49              # sample standard deviation
n_sample <- 80                # sample size
t = (xbar_sample-mu0)/(sample_s/sqrt(n_sample))    H0: miu = 11.5
t | > t                        # test statistic      H1: miu != 11.5
  [1] -10.15942

# We then compute the critical values at .05 significance level.

alpha <- .05
t.half.alpha <- qt(1-alpha/2, df=n_sample-1) > c(-t.half.alpha, t.half.alpha)
c(-t.half.alpha, t.half.alpha)              [1] -1.99045  1.99045
```

The test statistic -10.15942 does not lie between the critical values -1.99045 and 1.99045. Hence, at 0.05 significance level, we reject the null hypothesis that the average weekly study hours for students with a job is 11.5.





03



ANOVA

ANOVA ANALYSIS

```
# Compute the analysis of variance
res.aov <- aov (output_grade ~ sports_activity, data = project_anova)
#Summary of the analysis

summary(res.aov)                                > summary(res.aov)
|
|          Df Sum Sq Mean Sq F value Pr(>F)
| sports_activity    1     2.8   2.760    0.57  0.452
| Residuals      143   692.7   4.844
```

For the ANOVA, we analyzed if sport has a significant influence on students' output grade.

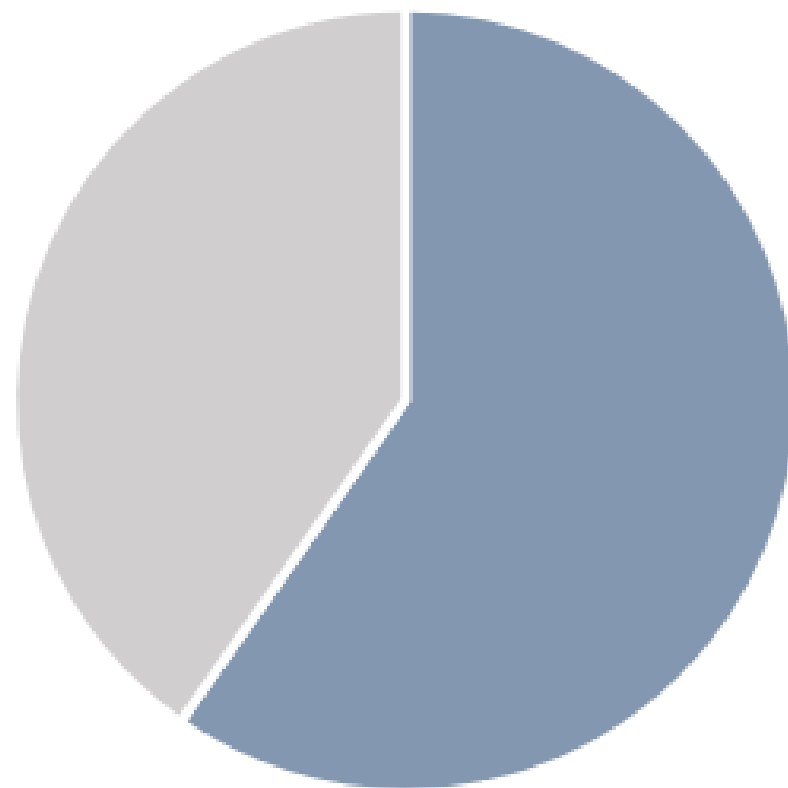
Based on the ANOVA results, there is no significant difference for what we analyzed, namely if sport has an impact on students' grades.

We took the confidence level of 95% and we observed that the ratio of between-group variation to within-group variance is shown by the F value of 0.57. Also corresponding p-value, is 0.452, beyond the usual 0.05 level of significance. As a result, we cannot rule out the null hypothesis.



GRAPHS

sports activity



■ NO ■ YES

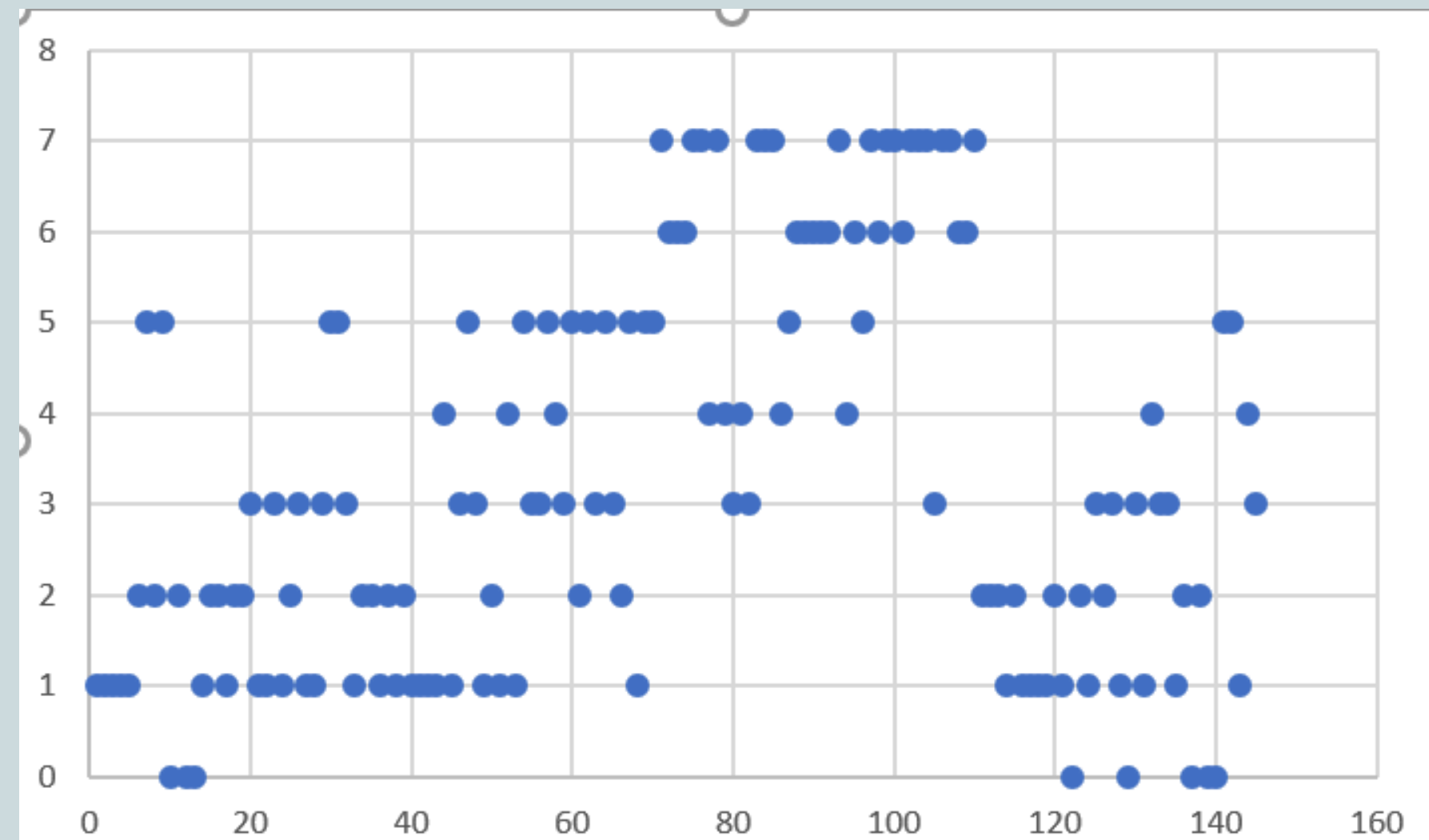
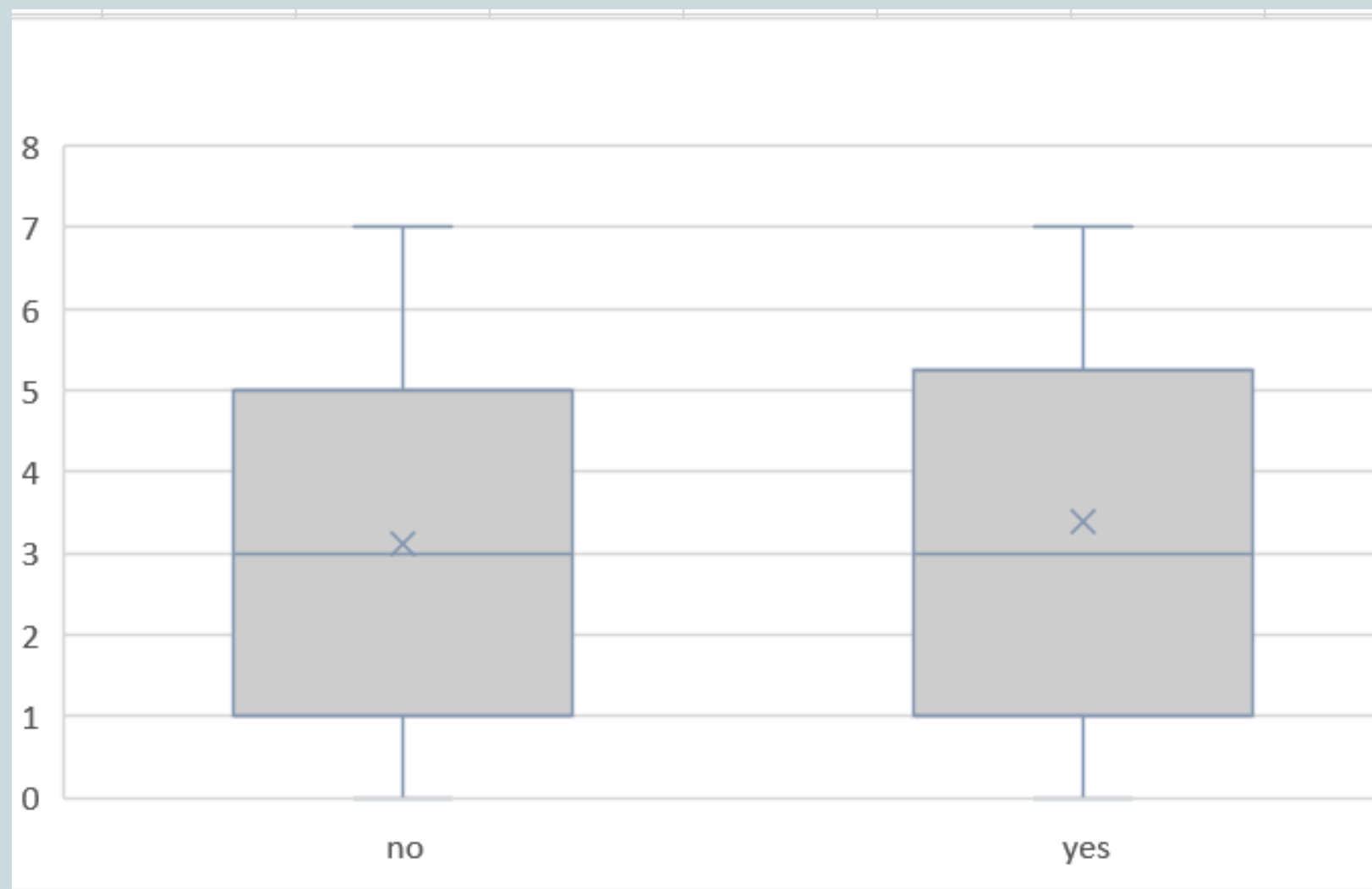
GRADES

1-poor
2-poor to satisfaction
3-satisfactory
4-satisfactory to good
5-good
6-good to excellent
7-excellent



■ 1 ■ 2 ■ 3 ■ 4 ■ 5 ■ 6 ■ 7

GRAPHS





04

SIMPLE LINEAR REGRESSION



LINEAR REGRESSION ANALYSIS



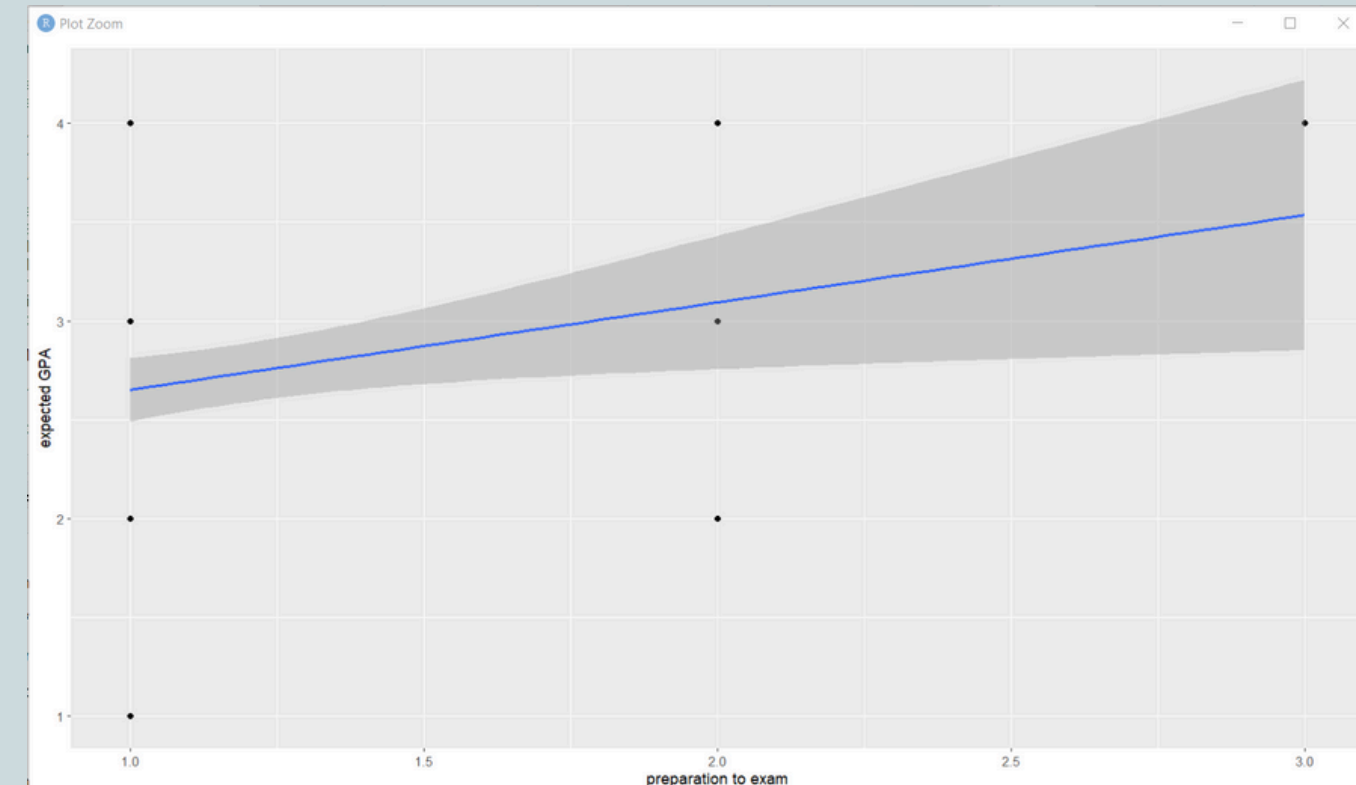
SIMPLE

```
Call:
lm(formula = `expected GPA` ~ `preparation to exam`, data = prep_expect)

Residuals:
    Min       1Q   Median       3Q      Max
-1.651 -0.651  0.349  0.349  1.349

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)      2.2090     0.2271   9.727  <2e-16 ***
`preparation to exam` 0.4420     0.1840   2.403  0.0175 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9017 on 143 degrees of freedom
Multiple R-squared:  0.03881,    Adjusted R-squared:  0.03209
F-statistic: 5.774 on 1 and 143 DF,  p-value: 0.01755
```



The estimated regression line equation can be written like this: $\text{expected_gpa} = 2.209 + 0.442 \cdot \text{prep_to_exam}$

In this linear regression model, the expected GPA is being predicted using the preparation to exam

The coefficient for the preparation to exam variable indicates that the expected change in GPA for each unit increases in the level of preparation to exam. In this case, the coefficient is 0.442, suggesting that, on average, for each unit increase in the level of preparation to exam, we expect a 0.442 increase in the expected GPA





LINEAR REGRESSION ANALYSIS

SIMPLE



```
> cor(pre_expect$`preparation to exam`, prep_expect$`expected GPA`)  
[1] 0.1970003
```

The positive correlation coefficient suggests that as the level of preparation for exams increases there is a slight tendency for the expected GPA to also increase

```
> confint(model)  
  
                2.5 %    97.5 %  
(Intercept)    1.76004065 2.6578698  
`preparation to exam` 0.07839748 0.8056439
```

Interpreting the confidence interval, we can say that for a one-unit increase in the preparation to exam variable, we would expect the variable expected GPA to increase by a value between 0.07839748 and 0.8056439, with 95% confidence.





05

MULTIPLE LINEAR REGRESSION



LINEAR REGRESSION ANALYSIS



MULTIPLE

```
Call:
lm(formula = `expected GPA` ~ `preparation to exam` + `attendance to classes` +
  `taking notes`, data = multiple_regr)

Residuals:
    Min       1Q   Median       3Q      Max
-1.8456 -0.5911  0.1544  0.6837  1.6633

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)      1.9902     0.4958   4.014 9.65e-05 ***
`preparation to exam`  0.3668     0.1842   1.991  0.0484 *
`attendance to classes` -0.2748     0.1768  -1.554  0.1224
`taking notes`      0.2545     0.1322   1.925  0.0562 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8861 on 141 degrees of freedom
Multiple R-squared:  0.08472,    Adjusted R-squared:  0.06525
F-statistic: 4.351 on 3 and 141 DF,  p-value: 0.005782
```

```
> summary(model_m)$coefficient
              Estimate Std. Error  t value    Pr(>|t|)
(Intercept)  1.9902119  0.4958069  4.014087 9.653204e-05
`preparation to exam`  0.3668231  0.1842123  1.991306 4.837945e-02
`attendance to classes` -0.2748145  0.1768151 -1.554248 1.223670e-01
`taking notes`  0.2544519  0.1321745  1.925122 5.622754e-02
```

$$\text{expected_gpa} = b_0 + b_1 \cdot \text{prep_to_exam} + b_2 \cdot \text{attend_class} + b_3 \cdot \text{taking_notes} + \text{epsilon}$$

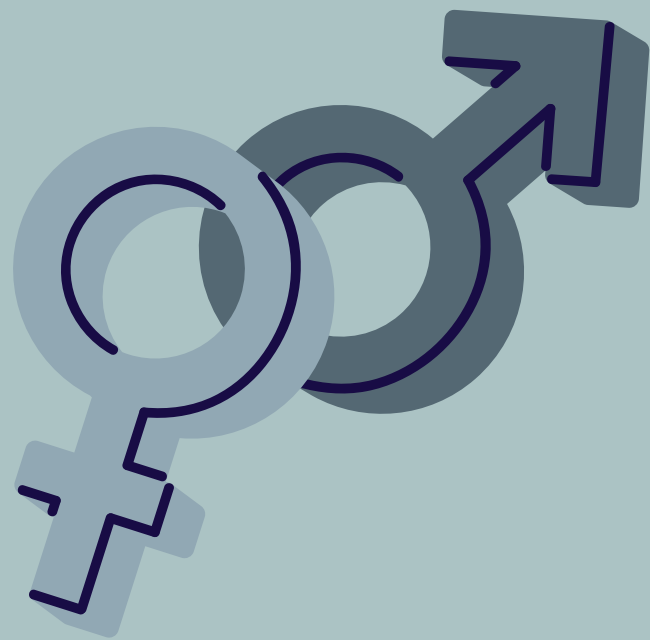
The Adjusted R-squared value for this model is 0.06525, which indicates that the predictor variables explain about 6.5% of the variability in the expected GPA.

Overall, based on this multiple linear regression model, it appears that preparation to exam has a statistically significant positive association with expected GPA, while the associations of attendance to classes and taking notes with expected GPA may not be statistically significant.



A value of 0.3252884 suggests that the standard deviation of the residuals is approximately 32.53% of the mean expected GPA. This indicates a moderate level of dispersion or variability in the expected GPA values around the predicted values from the regression model.

In other words, the expected GPA values are, on average, expected to deviate by about 32.53% from the predicted values provided by the regression model



The coefficient for the dummy variable gender (-0.5197) represents the difference in the expected GPA between females (coded as 1) and males (coded as 0) when all other variables are held constant. In this case, it suggests that the expected GPA for females is lower by 0.5197 compared to males, while considering the effects of the other variables

Comparing the two models, we can see that the adjusted R-squared has increased when the dummy variable was included. This suggests that the the gender variable as a predictor has improved the model's ability to explain the variation in the expected GPA

Dummy Variable

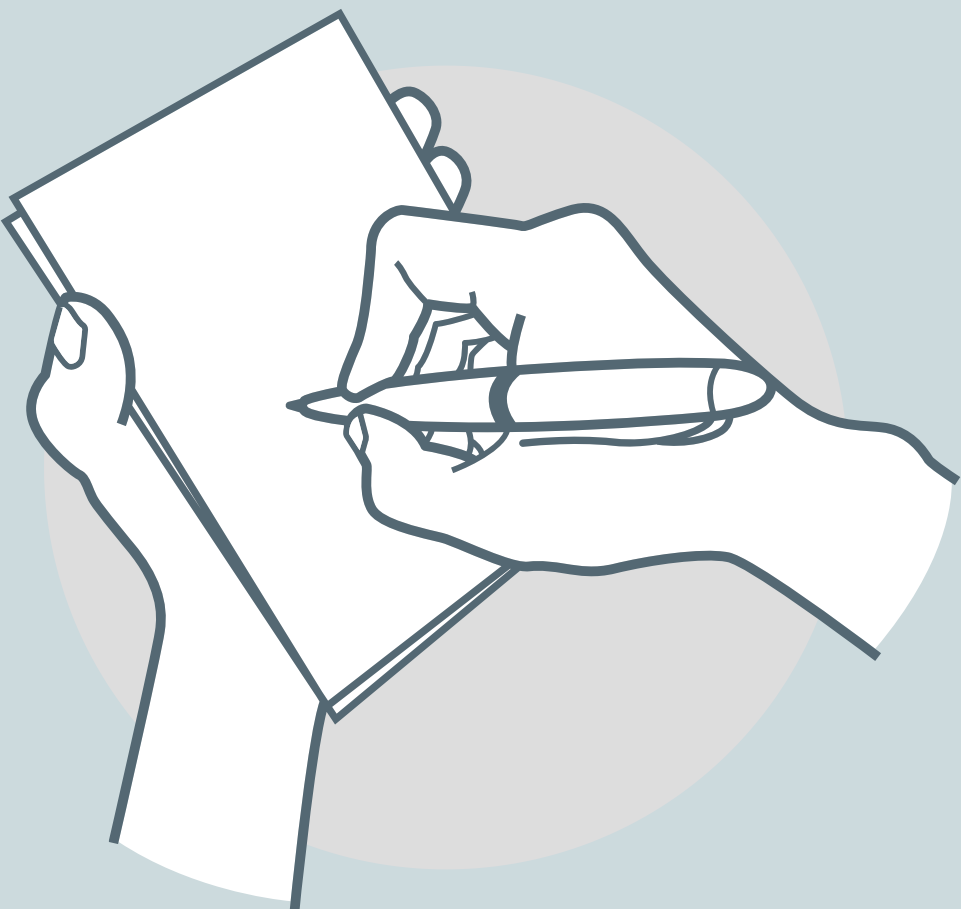
```
Call:
lm(formula = `expected GPA` ~ `preparation to exam` + `attendance to classes` +
  `taking notes` + dummy, data = multiple_dummy)

Residuals:
    Min       1Q   Median       3Q      Max
-2.02955 -0.50985 -0.02955  0.71857  1.49015

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)      2.1966     0.4796   4.580 1.02e-05 ***
`preparation to exam`  0.3850     0.1770   2.175 0.031274 *
`attendance to classes` -0.2374     0.1701  -1.395 0.165197
`taking notes`    0.2284     0.1271   1.797 0.074568 .
dummy           -0.5197     0.1449  -3.587 0.000461 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.851 on 140 degrees of freedom
Multiple R-squared:  0.1618,    Adjusted R-squared:  0.1378
F-statistic: 6.755 on 4 and 140 DF,  p-value: 5.32e-05
```

Interaction Terms



```
Call:
lm(formula = `expected GPA` ~ `taking notes` * `listening in classes` +
    dummy2, data = listening)

Residuals:
    Min       1Q   Median       3Q      Max
-2.19329 -0.53105 -0.08222  0.58002  1.54536

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)         0.3892     1.0313   0.377  0.706470
`taking notes`        0.9717     0.3994   2.433  0.016232 *
`listening in classes` 0.8758     0.4588   1.909  0.058310 .
dummy2              -0.5512     0.1495  -3.688  0.000323 ***
`taking notes`:`listening in classes` -0.3290     0.1777  -1.851  0.066219 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8641 on 140 degrees of freedom
Multiple R-squared:  0.1359,    Adjusted R-squared:  0.1112
F-statistic: 5.505 on 4 and 140 DF,  p-value: 0.0003811
```

The interaction term taking notes:listening in classes has a coefficient of -0.3290. This indicates that the combined effect of taking notes and listening in classes on expected GPA is negative, but the effect is not statistically significant at the conventional level (p-value = 0.066219).

Based on this model, taking better notes is associated with a higher expected GPA, while actively listening in classes shows a positive but non-significant relationship with expected GPA. Additionally, being female is associated with a lower expected GPA compared to being male, taking into account the other variables in the model. The interaction between taking notes and listening in classes does not have a statistically significant effect on the expected GPA

Poisson distribution

```
Call:
glm(formula = Y ~ X, family = poisson())

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.4879     0.2267   2.152  0.0314 *
X1           0.2065     0.1054   1.960  0.0500 .
X2           0.1518     0.1140   1.331  0.1832
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 48.461  on 144  degrees of freedom
Residual deviance: 42.908  on 142  degrees of freedom
AIC: 460.7

Number of Fisher Scoring iterations: 4
```

X1: the coefficient estimate for the predictor variable GENDER is 0.2065. However, the p-value (0.0500) suggests that the effect of gender on the expected GPA may not be statistically significant at the conventional significance level of 0.05.

X2: the coefficient estimate for the predictor variable PREP_EXAM is 0.1518. The p-value (0.1832) indicates that the effect of preparation to exam on the expected GPA is not statistically significant at the conventional significance level.

Overall, the model suggests that the intercept (representing the reference category) has a significant effect on the expected GPA

Overdispersion test

```
data: poisson_model
z = -16.69, p-value = 1
alternative hypothesis: true alpha is greater than 0
sample estimates:
    alpha
-0.2651946
```

The test result provides a z-value (-16.69) and a p-value (1), indicating strong evidence of overdispersion.

Negative binomial

```
Call:
glm.nb(formula = Y ~ X, init.theta = 178889.8854, link = log)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)   0.4879      0.2267   2.152   0.0314 *
X1             0.2065      0.1054   1.960   0.0500 .
X2             0.1518      0.1140   1.331   0.1832
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Negative Binomial(178889.9) family taken
as 1)

    Null deviance: 48.460  on 144  degrees of freedom
Residual deviance: 42.908  on 142  degrees of freedom
AIC: 462.71

Number of Fisher Scoring iterations: 1

              Theta: 178890
            Std. Err.: 2420896
Warning while fitting theta: iteration limit reached

2 x log-likelihood: -454.706
```

X1: The estimate of 0.2065 suggests that, on average, individuals with a higher value of GENDER x are associated with a 20.65% increase in the expected count of the response variable.

The p-value (0.05) indicates a marginally significant effect.

X2: The estimate of 0.1518 suggests that, on average, a one-unit increase in preparation for exam is associated with a 15.18% increase in the expected count of the response variable.

The p-value (0.1832) suggests that the effect may not be statistically significant at the usual significance level.

In our case, the estimated theta is 178890, indicating that the data has a high level of dispersion.

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**THANK
YOU**

